Helix Harbor Technologies

Internal Knowledge Base - RAG / Vector DB Test Document

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1. Company Overview

Helix Harbor Technologies was founded in 2011 in Rotterdam, Netherlands by Marit Van der Linde and Joaquin Okafor. The company builds low-power environmental sensors for ports, fish farms, and offshore wind installations. Its headquarters moved to Lisbon, Portugal in 2019, and it now employs 412 people across 9 offices.

The company reported annual revenue of 87.4 million euros in fiscal year 2025, a 23 percent increase over the 71.0 million euros recorded in 2024. The current Chief Executive Officer is Dr. Ingrid Solheim, who succeeded Marit Van der Linde in January 2023. The Chief Technology Officer is Tomas Beaumont.

Helix Harbor operates three regional data centers: Frankfurt (EU-CENTRAL), Singapore (APAC-SOUTH), and Virginia (US-EAST). The Frankfurt facility is the primary site and handles 61 percent of all sensor traffic.

2. Product Catalog

The flagship product line is AquaSense, a family of submersible sensors that measure salinity, dissolved oxygen, temperature, and turbidity. Each unit is rated for continuous operation at depths up to 300 meters.

2.1 AquaSense Model Comparison

Model	Max Depth	Battery Life	Price (EUR)	Warranty
AquaSense Lite (AS-100)	50 m	9 months	340	1 year
AquaSense Pro (AS-300)	150 m	18 months	890	3 years
AquaSense Deep (AS-900)	300 m	30 months	2,150	5 years
AquaSense Array (AS-X)	300 m	36 months	7,400	5 years

The AS-900 Deep uses a titanium housing certified to IP69K and pressure-tested to 45 bar. The AS-100 Lite uses a polycarbonate housing and is not recommended for use in water with a pH below 5.5 or above 9.0.

2.2 Connectivity

All AquaSense models transmit data using LoRaWAN on the 868 MHz band in Europe and 915 MHz in the Americas. The AS-900 and AS-X additionally support satellite uplink through the Orbcomm network, with a default reporting interval of 15 minutes and a minimum interval of 60 seconds.

3. Returns and Refund Policy

Customers may return any unused AquaSense device within 30 days of delivery for a full refund, minus a restocking fee of 12 percent. Devices that have been submerged are considered used and are not eligible for a refund, but remain covered by warranty.

To start a return, customers must contact support at returns@helixharbor.example and provide the order number, which always begins with the prefix HH-ORD followed by eight digits. A return authorization (RMA) is issued within 2 business days. Refunds are processed within 10 business days after the returned device passes inspection.

Orders exceeding 25,000 euros are handled by the Enterprise Accounts team and are subject to a negotiated return window of up to 90 days. Custom-calibrated AS-X Array units are non-refundable once calibration has begun.

3.1 Warranty Claims

Warranty covers manufacturing defects but excludes damage from biofouling, lightning strikes, and physical impact. Claims require photographs of the device and its serial number, which starts with the characters HX followed by a 10-character alphanumeric code. Replacement units ship within 5 business days of claim approval.

4. Information Security Policy

All employees must enable multi-factor authentication using a hardware security key. SMS-based codes are prohibited. Passwords must be at least 16 characters long and rotated only when compromise is suspected, in line with NIST 800-63B guidance.

Customer sensor data is encrypted in transit with TLS 1.3 and at rest with AES-256-GCM. Encryption keys are rotated every 90 days and stored in a hardware security module located in the Frankfurt data center. Access to production databases requires approval from two engineers and is logged for 400 days.

4.1 Incident Response

Security incidents are classified into four severity levels, SEV1 through SEV4. A SEV1 incident requires the on-call engineer to acknowledge within 5 minutes and the incident commander to notify the CEO within 30 minutes. Customers must be notified of any confirmed data breach within 72 hours, as required by GDPR Article 33.

Severity	Definition	Response Time	Update Cadence
SEV1	Total outage or confirmed data breach	5 minutes	Every 30 minutes
SEV2	Major feature degraded for many customers	15 minutes	Every 1 hour
SEV3	Minor feature degraded, workaround exists	4 hours	Daily
SEV4	Cosmetic issue or documentation error	2 business days	On resolution

5. API Reference

The Helix Harbor REST API is available at https://api.helixharbor.example/v2. Authentication uses bearer tokens obtained from the /auth/token endpoint. Tokens expire after 3,600 seconds. The default rate limit is 600 requests per minute per API key, and exceeding it returns HTTP status 429.

5.1 Retrieve Latest Reading

```
GET /v2/devices/{device_id}/readings/latest
Authorization: Bearer <token>
```

Returns the most recent reading for the specified device. The response contains the fields temperature_c, salinity_psu, dissolved_oxygen_mgl, turbidity_ntu, and recorded_at in ISO 8601 format. If the device has not reported in the past 24 hours, the endpoint returns status 404 with the error code DEVICE_STALE.

5.2 Create Alert Rule

```
POST /v2/alerts
{"device_id": "HX7K2M9QP4", "metric": "dissolved_oxygen_mgl", "operator": "lt",
"threshold": 4.5}
```

Creates an alert that triggers when the metric crosses the threshold. Each account may define up to 250 alert rules. Alerts are delivered by email, webhook, or SMS, and webhook deliveries are retried 6 times with exponential backoff.

5.3 Error Codes

HTTP Status	Error Code	Meaning
400	INVALID_PARAMETER	A request field is missing or malformed
401	TOKEN_EXPIRED	Bearer token is expired or revoked
404	DEVICE_STALE	No reading in the last 24 hours
409	RULE_LIMIT_REACHED	Account has 250 alert rules already
429	RATE_LIMITED	More than 600 requests per minute

6. Employee Leave and Benefits

Full-time employees receive 27 days of paid annual leave plus public holidays in their country of employment. Unused leave of up to 5 days may be carried over into the following year and expires on 31 March. Parental leave is 18 weeks at full pay for all parents, regardless of gender.

Each employee receives an annual learning budget of 1,800 euros, which can be used for courses, conferences, or books. Remote work is permitted up to 3 days per week for roles that do not require on-site hardware access. Employees working from home receive a monthly stipend of 45 euros.

The company offsite takes place each September in Madeira. In 2026 it is scheduled for 21 to 24 September. Attendance is optional but encouraged, and travel and lodging are fully covered.

7. Frequently Asked Questions

Q: What is the maximum depth of the AquaSense Pro?

A: The AquaSense Pro (AS-300) is rated to 150 meters.

Q: Which frequency does LoRaWAN use in Europe?

A: The 868 MHz band is used in Europe.

Q: How long are encryption keys valid?

A: Keys are rotated every 90 days.

Q: Who founded Helix Harbor Technologies?

A: Marit Van der Linde and Joaquin Okafor founded the company in 2011 in Rotterdam.

Q: Can I return a device that has been submerged?

A: No. Submerged devices count as used and are not refundable, though warranty still applies.

Q: How many alert rules can one account create?

A: Up to 250 alert rules per account.

8. Appendix: Deployment Incident Log (2025)

The following table records notable incidents from 2025 and is intended to test table-aware chunking and numeric retrieval.

Date	ID	Severity	Region	Duration	Root Cause
2025-01-19	INC-0412	SEV2	EU-CENTRAL	47 min	Expired TLS certificate on ingest gateway
2025-03-07	INC-0438	SEV3	APAC-SOUTH	3 h 10 min	Misconfigured rate limiter
2025-05-23	INC-0467	SEV1	US-EAST	2 h 05 min	Database failover did not complete
2025-08-11	INC-0491	SEV2	EU-CENTRAL	1 h 22 min	Satellite uplink queue backlog
2025-10-30	INC-0522	SEV4	APAC-SOUTH	n/a	Typo in API documentation
2025-12-02	INC-0540	SEV2	US-EAST	58 min	Faulty deployment of alert service v4.8.1

Total downtime attributed to SEV1 incidents in 2025 was 2 hours and 5 minutes, all in the US-EAST region. The longest SEV2 incident was INC-0491 at 1 hour and 22 minutes.